

Random experiment:

An experiment whose all possible outcomes are known, but it is not possible to predict the outcome.

Ex:

i) A fair coin is tossed

ii) A die is rolled are random

Experiments, Since we cannot predict the outcome of the experiment in any trial.

Sample Space:

The set of all possible outcomes of a random experiment is called a Sample Space.

Need of Random variables:

In nature, the outcomes of random experiments may be numerical or non-numerical.

For example,

When we throw a die, the outcomes are numerical. (we get the outcomes as 1, 2, 3, 4, 5 or 6).

But when we toss a coin, the outcomes are non-numerical or descriptive. (we get the outcomes as head or tail).

It is inconvenient to deal with these descriptive outcomes mathematically. We are always interested and feel convenient in dealing with numerical outcomes, whether the experiments results in numerical or non-numerical outcomes.

Hence we may assign a real number to each of the outcomes using a fixed rule or mapping.

For example, when we toss a coin we get two outcomes, namely head or tail. Instead of dealing the outcomes descriptively as head or tail; we can assign numerical values, say '1' to head and '0' to tail. This interpretation is

easy and attractive from mathematical point of view and also practically meaningful.

Random variable:

A random variable is a function from sample space (S) to set of all real numbers (R).
ie $X: S \rightarrow R$.

Ex:

Let us consider the tossing of two fair coins at a time.

The possible outcomes are $\{HH, HT, TH, TT\}$.

Let us consider the variable X which is "the number of heads obtained" while tossing two fair coins.

we could assign the value $X=0$ to the outcome of getting no heads, $X=1$ to the outcome of getting only 1 head and $X=2$ to the outcome of getting 2 heads.

$$\therefore X(TT) = 0, X(TH) = 1,$$

$$X(HT) = 1, X(HH) = 2.$$

$\therefore X$ takes the values 0, 1, 2.

Thus we can assign a real number $X(s)$ to every element s of the sample space S .

A random variable is also called a chance variable or a stochastic variable.

Types of Random Variables:

1. Discrete Random Variable
2. Continuous Random Variable

Discrete Random Variable:

If X is a random variable which takes a finite number (or) countably infinite number of values then X is called a discrete random variable.

Ex:

1. The number of heads obtained when two coins are tossed is a discrete random variable as X assumes the values 0, 1 or 2 which form a countable set.

2. The number of alpha particles emitted by a radio active source.

Notation:

Consider an experiment of tossing an unbiased coin twice.

Let the random variable X denote the number of heads turning up.

The outcome of the experiments are HH, HT, TH and TT.

Hence X can take values 2, 1, 1, 0.

Here $P(X \leq 1)$ means $P(\text{the number of heads} \leq 1)$

$$= P(HT, TH, TT)$$

$$P(X \leq 1) = \frac{3}{4}$$

Generally,

$P(X \leq a)$ means that the probability of the set of outcomes ' ω ' in the sample space for which $X(\omega) \leq a$ (or)

$$\begin{aligned} P(X \leq a) &= P(\{\omega : X(\omega) \leq a\}) \\ &= P(\{\omega : X(\omega) \in [-\infty, a]\}) \end{aligned}$$

Note:

x be a real number

1. Let 'S' be a sample space and The set of all outcomes 's' in S is such that $X(s) = x$ is denoted by writing $X = x$.

$$\text{ie } P(X = x) = P(\{s : X(s) = x\})$$

$$2. P(a < X \leq b) = P(\{s : X(s) \in (a, b]\})$$

$$3. P(X \geq a) = 1 - P(X < a)$$

$$4. P(X \leq a) = 1 - P(X > a)$$

Theorem:

If X_1 and X_2 are random Variables and k is a Constant then kX_1 , $X_1 + X_2$, $X_1 X_2$, $k_1 X_1 + k_2 X_2$, $X_1 - X_2$ are also random variables.

one-dimensional random variable
(or) Scalar-valued random variable:

If a random Variable X takes on single value corresponding to each outcome of the experiment, then the random variable is called one-dimensional random variable.

Ex:

In coin tossing experiment, if

we assume the random variable to be appearance of tail, then the sample space is $\{H, T\}$ and the random variable is $\{0, 1\}$, which is an one-dimensional random variable.

Probability Mass function (or)
probability function:

Let X be a discrete random variable which takes the values $x_1, x_2, x_3, \dots$ (countably infinite number of values).

Then the discrete probability function $p(x_i)$ or $p(X=x_i) = p_i$ or p_i satisfies the following conditions

$p(x)$ $P(X=x)$ p_i

$$i) p(x_i) \geq 0 \quad (\text{or}) p_i \geq 0 \text{ for all } i$$

$$ii) \sum_{i=1}^{\infty} p(x_i) = 1 \quad \sum p_i = 1$$

is called probability mass function or probability function or point probability function.

Distribution function of the random variable X (or) Cumulative distribution function of the random variable X:

Distribution function of a random variable X defined in $(-\infty, \infty)$ is given by

$$\begin{aligned} F(x) &= P(X \leq x) = P(\{s: X(s) \leq x\}) \\ &= \sum_{x_i \leq x} p(x_i) \end{aligned}$$

Properties:

$$1. P(a < X \leq b) = F(b) - F(a)$$

where $F(x) = P(X \leq x)$

$$2. P(a \leq X \leq b) = F(b) - F(a) + P(X=a)$$

$$3. P(a < X < b) = F(b) - F(a) - P(X=b)$$

$$4. P(a \leq X < b) = F(b) - F(a) - P(X=b) + P(X=a)$$

Also

$$i) 0 \leq F(x) \leq 1$$

$$ii) \text{ If } x < y \text{ then } F(x) \leq F(y)$$

$$iii) F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = 0$$

$$iv) F(\infty) = \lim_{x \rightarrow \infty} F(x) = 1$$

Example:

① Find the probability mass function and cumulative distribution function for getting number of heads when three coins are tossed once.

Sol:

Let X be a random variable "getting number of heads".

Sample space when three coins are tossed is

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

$S:$	HHH	HHT	HTH	THH	HTT	THT	TTH	TTT
P (No. of heads)	3	2	2	2	1	1	1	0

Since X is a random variable getting the number of heads, X takes the values $0, 1, 2$ & 3 . ($X: S \rightarrow R$)

$$P(\text{getting no head}) = P(X=0) = \frac{1}{8}$$

$$P(\text{getting one head}) = P(X=1) = \frac{3}{8}$$

$$P(\text{getting two heads}) = P(X=2) = \frac{3}{8}$$

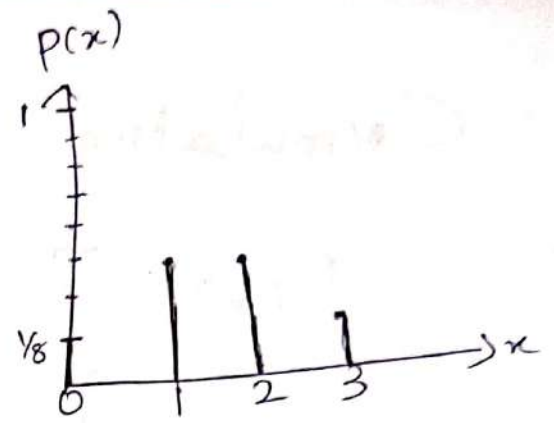
$$P(\text{getting three heads}) = P(X=3) = \frac{1}{8}$$

$\therefore$ Probability mass function is given

$$\text{by } P(X=x) = \begin{cases} \frac{1}{8} & \text{if } x=0 \\ \frac{3}{8} & \text{if } x=1 \\ \frac{3}{8} & \text{if } x=2 \\ \frac{1}{8} & \text{if } x=3 \end{cases}$$

(or)

$X :$	0	1	2	3
$P(X=x) :$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$



Cumulative distribution function:

When $x=0$, $F(0) = P(X=0) = \frac{1}{8}$

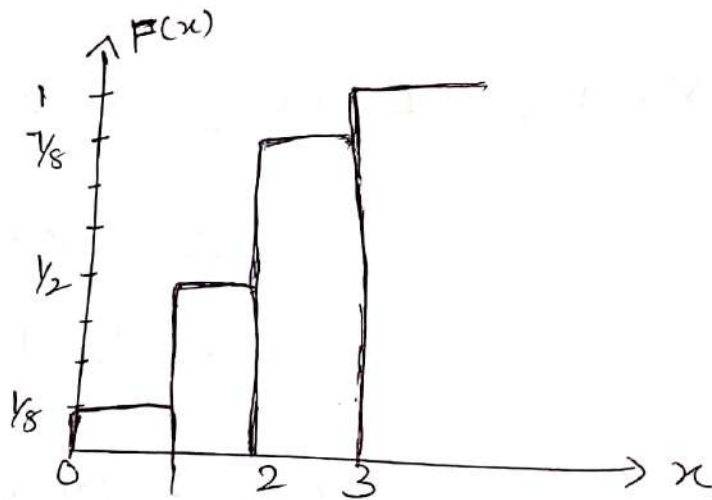
When $x=1$, $F(1) = P(X \leq 1)$
 $= P(X=0) + P(X=1)$
 $= \frac{1}{8} + \frac{3}{8} = \frac{4}{8} = \frac{1}{2}$

When $x=2$, $F(2) = P(X \leq 2)$
 $= P(X=0) + P(X=1) + P(X=2)$
 $= \frac{1}{8} + \frac{3}{8} + \frac{3}{8} = \frac{7}{8}$

When $x=3$, $F(3) = P(X \leq 3)$
 $= P(X=0) + P(X=1) + P(X=2) + P(X=3)$
 $= \frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8}$
 $= 1$

Cumulative distribution function is

$$F(x) = \begin{cases} 0 & \text{if } -\infty < x < 0 \\ 1/8 & \text{if } 0 \leq x < 1 \\ 1/2 & \text{if } 1 \leq x < 2 \\ 7/8 & \text{if } 2 \leq x < 3 \\ 1 & \text{if } 3 \leq x < \infty \end{cases}$$



② Find the probability mass function and the cumulative distribution function for getting '3' & when two dice are thrown.

Sol:

Two dice are thrown. Let X be the random variable of getting number of '3' &. Therefore X can take the values 0, 1, 2.

Sample space is

$$\left\{ \begin{array}{l} (1,1), (1,2), (1,3), (1,4), (1,5), (1,6) \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6) \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6) \\ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6) \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6) \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6) \end{array} \right\}$$

$$P(\text{no '3'}) = P(X=0) = \frac{25}{36}$$

$$P(\text{one '3'}) = P(X=1) = \frac{10}{36}$$

$$P(\text{two '3's'}) = P(X=2) = \frac{1}{36}$$

Probability mass function is given by

x	0	1	2
$P(X=x)$	$\frac{25}{36}$	$\frac{10}{36}$	$\frac{1}{36}$

Cumulative distribution function:

$$F(0) = P(X \leq 0) = \frac{25}{36}$$

$$\begin{aligned} F(1) &= P(X \leq 1) = P(X=0) + P(X=1) \\ &= \frac{25}{36} + \frac{10}{36} = \frac{35}{36} \end{aligned}$$

$$\begin{aligned} F(2) &= P(X \leq 2) = P(X=0) + P(X=1) + P(X=2) \\ &= \frac{25}{36} + \frac{10}{36} + \frac{1}{36} \\ &= \frac{36}{36} = 1. \end{aligned}$$

x	0	1	2
$F(x)$	$\frac{25}{36}$	$\frac{35}{36}$	1

③ A random variable X has the following probability mass function

$$x: 0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6$$

$$P(X=x): k \quad 3k \quad 5k \quad 7k \quad 9k \quad 11k \quad 13k$$

① Find k

② Evaluate $P(X < 4)$, $P(X \geq 5)$ and $P(3 < X \leq 6)$.

③ What is the smallest value of x for which $P(X \leq x) > \frac{1}{2}$.

Sol:

① Since $P(X=x)$ is a probability mass function, $\sum_{x=0}^6 P(X=x) = 1$.

$$\text{c.e. } P(x=0) + P(x=1) + P(x=2) + P(x=3) + P(x=4) + P(x=5) + P(x=6) = 1$$

$$\Rightarrow k + 3k + 5k + 7k + 9k + 11k + 13k = 1$$

$$\Rightarrow 49k = 1$$

$$\Rightarrow \boxed{k = \frac{1}{49}}$$

$$\begin{aligned} \textcircled{2} \quad P(x < 4) &= P(x=0) + P(x=1) + P(x=2) + P(x=3) \\ &= \frac{1}{49} + \frac{3}{49} + \frac{5}{49} + \frac{7}{49} = \frac{16}{49} \end{aligned}$$

$$P(x \geq 5) = P(x=5) + P(x=6) = \frac{11}{49} + \frac{13}{49} = \frac{24}{49}$$

$$\begin{aligned} P(3 < x \leq 6) &= P(x=4) + P(x=5) + P(x=6) \\ &= \frac{9}{49} + \frac{11}{49} + \frac{13}{49} = \frac{33}{49} \end{aligned}$$

③ The minimum value of x may be determined by trial and error method.

$$P(X \leq 0) = \frac{1}{49} < \frac{1}{2}$$

$$P(X \leq 1) = \frac{4}{49} < \frac{1}{2}$$

$$P(X \leq 2) = \frac{9}{49} < \frac{1}{2}$$

$$P(X \leq 3) = \frac{16}{49} < \frac{1}{2}$$

$$P(X \leq 4) = \frac{25}{49} > \frac{1}{2}$$

$\therefore$ The smallest value of x for which $P(X \leq x) > \frac{1}{2}$ is 4.

Continuous Random Variable:

A random variable X is said to be continuous if it can take all possible values between certain given limits.

Ex:

1. age, height, weight.
2. The life length in hours of a certain light bulb.

Probability density function (pdf)
(or) Continuous probability function:

The function $f(x)$ is a probability density function for a

Continuous random variable X , defined over the set of real numbers, if

1. $f(x) \geq 0$ for all $x \in \mathbb{R}$

2. $\int_{-\infty}^{\infty} f(x) dx = 1$

3. $P(a < X < b) = \int_a^b f(x) dx$

Note:

Since continuous probability function are defined for uncountable number of points over an interval, the probability at a single point is always zero.

i.e. $P(X=a) = \int_a^a f(x) dx = 0$

The probabilities are measured over intervals and not at single points. That is, the area under the

Curve between two distinct points defines the probability for that interval.

$$\therefore P(a \leq x \leq b) = P(a \leq x < b) = P(a < x \leq b) = P(a < x < b).$$

i.e. In case of continuous random variable, it does not matter whether we include or not the end points of the interval from a to b .

Cumulative Distribution Function
(cdf) :

If X is a continuous random variable, then the function given by

$$F(x) = P(-\infty < X \leq x) = \int_{-\infty}^x f(x) dx$$

is called the distribution function
(or) Cumulative distribution of X .

Properties of distribution function:

1. $F(x)$ is a non-decreasing function of x .

2. $0 \leq F(x) \leq 1$, $-\infty < x < \infty$.

3. $F(-\infty) = \lim_{x \rightarrow -\infty} \int_{-\infty}^x f(x) dx = \int_{-\infty}^{-\infty} f(x) dx = 0$.

4. $F(\infty) = \lim_{x \rightarrow \infty} \int_{-\infty}^x f(x) dx = \int_{-\infty}^{\infty} f(x) dx = 1$.

5. For any real constants a and b and $a \leq b$, $P(a \leq X \leq b) = F(b) - F(a)$.

$$6. f(x) = \frac{d}{dx} (F(x))$$

$$\text{ie } F'(x) = f(x) \rightarrow$$

① A Continuous random variable x follows the probability law,

$$f(x) = \begin{cases} kx(1-x)^{10}, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find k .

Sol:

Since $f(x)$ is a p.d.f,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\text{ie } \int_0^1 kx(1-x)^{10} dx = 1$$

$$\Rightarrow \int_0^1 k(1-x)(1-(1-x))^{10} dx = 1$$

[By properties of definite integral

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx \quad (ii)$$

$$\Rightarrow k \int_0^1 x^{10} (1-x) dx = 1$$

$$\Rightarrow k \int_0^1 (x^{10} - x^{11}) dx = 1$$

$$\Rightarrow k \left[\frac{x^{11}}{11} - \frac{x^{12}}{12} \right]_0^1 = 1$$

$$\Rightarrow k \left[\frac{1}{11} - \frac{1}{12} \right] = 1$$

$$\Rightarrow k \left[\frac{1}{132} \right] = 1$$

$$\Rightarrow \boxed{k = 132}$$

② A Continuous random variable X has pdf $f(x) = 3x^2$, $0 \leq x \leq 1$.

Find a and b such that

$$(i) P(x \leq a) = P(x > a) \text{ and}$$

$$(ii) P(x > b) = 0.05.$$

Sol:

(i) Since the total probability is 1,

$$\left[\text{Given that } P(x \leq a) = P(x > a) \right]$$

$$P(x \leq a) + P(x > a) = 1$$

$$\text{ie } P(x \leq a) + P(x \leq a) = 1$$

$$\Rightarrow P(x \leq a) = \frac{1}{2}$$

$$\Rightarrow \int_0^a f(x) dx = \frac{1}{2}$$

$$\Rightarrow \int_0^a 3x^2 dx = \frac{1}{2}$$

$$\Rightarrow \left(\frac{3x^3}{3} \right)_0^a = \frac{1}{2}$$

$$\Rightarrow a^3 = \frac{1}{2}$$

$$\Rightarrow a = \left(\frac{1}{2} \right)^{\frac{1}{3}}$$

$$(ii) P(X > b) = 0.05$$

$$\therefore \int_b^1 f(x) dx = 0.05$$

$$\therefore \int_b^1 3x^2 dx = 0.05$$

$$\Rightarrow \left(\frac{3x^3}{3} \right)_b^1 = 0.05$$

$$\Rightarrow 1 - b^3 = 0.05$$

$$\Rightarrow 1 - 0.05 = b^3$$

$$\Rightarrow b^3 = 0.95$$

$$\Rightarrow b^3 = \frac{95}{100}$$

$$\Rightarrow b^3 = \frac{19}{20}$$

$$\Rightarrow b = \left(\frac{19}{20} \right)^{1/3}$$

③ If the probability density function of a random variable is given by $f(x) = \begin{cases} k(1-x^2), & 0 \leq x < 1 \\ 0, & \text{elsewhere} \end{cases}$ find (i) k (ii) the distribution function of the random variable.

Sol:

(i) Since $f(x)$ is a p.d.f,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_0^1 k(1-x^2) dx = 1$$

$$\Rightarrow k \left(x - \frac{x^3}{3} \right) \Big|_0^1 = 1$$

$$\Rightarrow k \left(1 - \frac{1}{3} \right) = 1$$

$$\Rightarrow \frac{2}{3} k = 1$$

$$\Rightarrow \boxed{k = \frac{3}{2}}$$

(ii) The distribution function $F(x) = \int_{-\infty}^x f(x) dx$

(a) When $x \in (-\infty, 0]$

$$F(x) = \int_{-\infty}^x f(x) dx = 0$$

(b) When $x \in (0, 1)$

$$F(x) = \int_{-\infty}^x f(x) dx$$

$$= \int_{-\infty}^0 f(x) dx + \int_0^x f(x) dx$$

$$= 0 + \int_0^x \frac{3}{2} (1-x^2) dx$$

$$= \frac{3}{2} \left(x - \frac{x^3}{3} \right)_0^x$$

$$= \frac{3}{2} \left(x - \frac{x^3}{3} \right)$$

(c) When $x \in [1, \infty)$

$$F(x) = \int_{-\infty}^x f(x) dx$$

$$= \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^x f(x) dx$$

$$= 0 + \int_0^1 \frac{3}{2} (1-x^2) dx + 0$$

$$= \frac{3}{2} \left(x - \frac{x^3}{3} \right) \Big|_0^1$$

$$= \frac{3}{2} \left(1 - \frac{1}{3} \right)$$

$$= \frac{3}{2} \left(\frac{2}{3} \right)$$

$$= 1.$$

$$\therefore F(x) = \begin{cases} 0 & -\infty < x \leq 0 \\ \frac{3}{2} \left(x - \frac{x^3}{3} \right) & 0 < x < 1 \\ 1 & 1 \leq x < \infty \end{cases}$$

④ If $F(x) = \frac{1}{\pi} \left(\frac{\pi}{2} + \tan^{-1} x \right)$, $-\infty < x < \infty$ is a distribution function of a continuous variable X , find $P(0 \leq X \leq 1)$.

Sol: $F(x) = \frac{1}{\pi} \left(\frac{\pi}{2} + \tan^{-1} x \right)$

$$P(0 \leq X \leq 1) = F(1) - F(0)$$

$$= \frac{1}{\pi} (\pi/2 + \tan^{-1} 1) - \frac{1}{\pi} (\pi/2 + \tan^{-1} 0)$$

$$= \frac{1}{\pi} (\pi/2 + \pi/4) - \frac{1}{\pi} (\pi/2 + 0)$$

$$= \frac{1}{\pi} (\cancel{\pi/2} + \pi/4 - \cancel{\pi/2})$$

$$P(0 \leq x \leq 1) = \frac{1}{4}$$

⑤ If $f(x) = \begin{cases} A/x, & 1 < x < e^3 \\ 0, & \text{elsewhere} \end{cases}$

is a probability density function of a Continuous random Variable X , find $P(x > e)$.

Sol: Since $f(x)$ is a p.d.f,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_1^{e^3} A/x dx = 1$$

$$\Rightarrow A (\log x)_1^{e^3} = 1$$

$$\Rightarrow A (\log e^3 - \log 1) = 1$$

$$\Rightarrow A (3 - 0) = 1$$

$$\Rightarrow \boxed{A = \frac{1}{3}}$$

$$\boxed{\log 1 = 0}$$

$$P(x > e) = \frac{1}{3} \int_e^{e^3} \frac{1}{x} dx$$

$$= \frac{1}{3} (\log x)_e^{e^3}$$

$$= \frac{1}{3} (\log e^3 - \log e)$$

$$= \frac{1}{3} (3 - 1) = \frac{2}{3}$$

$$\boxed{P(x > e) = \frac{2}{3}}$$

⑥ For the probability density function

$$f(x) = \begin{cases} 2e^{-2x}, & x > 0 \\ 0, & x \leq 0 \end{cases} \quad \text{find } F(2).$$

Sol:

$$F(2) = P(X \leq 2) = \int_{-\infty}^2 f(x) dx$$

$$= \int_{-\infty}^0 f(x) dx + \int_0^2 f(x) dx$$

$$= 0 + \int_0^2 2e^{-2x} dx$$

$$= 0 + \left(\frac{e^{-2x}}{-1} \right)_0^2$$

$$= -e^{-4} + 1$$

$$= 1 - e^{-4}$$

$$= 1 - \frac{1}{e^4}$$

$$= \frac{e^4 - 1}{e^4}$$

$$\because e^0 = 1$$

⑦ The total life time (in year) of 5 year old dog of a certain breed is a random variable whose distribution function is given by

$$F(x) = \begin{cases} 0, & \text{for } x \leq 5 \\ 1 - \frac{25}{x^2}, & \text{for } x > 5 \end{cases}$$

Find the probability that such a five year old dog will live

- (i) beyond 10 years
- (ii) less than 8 years
- (iii) anywhere between 12 to 15 years.

Sol:

(1) $P(\text{dog living beyond 10 years})$

$$P(X > 10) = 1 - P(X \leq 10)$$

$$= 1 - \left(1 - \frac{25}{x^2}\right)$$

When $x=10$

$$= 1 - \left(1 - \frac{25}{100}\right)$$

$$= 1 - \left(1 - \frac{1}{4}\right)$$

$$= 1 - \frac{3}{4}$$

$$= \frac{1}{4}$$

(ii) $P(\text{dog living less than 8 years})$

$$P(X < 8) = F(8)$$

$$= 1 - \frac{25}{8^2}$$

$$= 1 - \frac{25}{64}$$

$$= \frac{39}{64}$$

[Since $P(X < 8) = P(X \leq 8)$ for a Continuous distribution]

(iii) $P(\text{dog living any where between 12 to 15 years}) = P(12 < X < 15)$

$$= F(15) - F(12)$$

$$= \left(1 - \frac{25}{15^2}\right) - \left(1 - \frac{25}{12^2}\right)$$

$$= \frac{1}{16}$$

Expected value is the average value of a random variable over a large number of experiments.

Expected Value:

Expected value is a predicted value of a variable, calculated as the sum of all possible values each multiplied by the probability of its occurrence.

Mathematical Expectation: (Expected value)

Expectation of a discrete random Variable

If X denote a discrete random Variable which can assume the values $x_1, x_2, \dots, x_n$ with respective probabilities $p_1, p_2, \dots, p_n$ then the mathematical expectation of X , denoted by $E(X)$ and is defined by

$$E(X) = p_1 x_1 + p_2 x_2 + \dots + p_n x_n$$

$$= \sum_{i=1}^n p_i x_i \quad \text{where } \sum_{i=1}^n p_i = 1.$$

Thus $E(X)$ is the weighted arithmetic mean of the values x_i with the weights $p(x_i)$.

$$\therefore \overline{X} = E(X)$$

Hence the mathematical Expectation $E(X)$ of a random Variable is simply arithmetic mean.

Properties:

1. $E(c) = c$ where c is a constant.
2. $E(cX) = cE(X)$
3. $E(aX + b) = aE(X) + b$.

Moments:

Expected values of a function of a random variable X is used for calculating the moments.

Types of Moments:

- (i) Moments about the origin.
- (ii) Moments about the mean which are called central moments.

Moments about the origin:

If X is a discrete random variable for each positive integer r

($r = 1, 2, \dots$) the r^{th} moment

$$\mu_r' = \sum p_i x_i^r = E(X^r).$$

First Moment:

$$\mu_1' = E(X) = \sum p_i x_i$$

This is called the mean of the random variable X .

Second Moment:

$$\mu_2' = E(X^2) = \sum p_i x_i^2$$

Note:

r^{th} moment about the point A :

$$\mu_r' = \sum_i (x_i - A)^r p_i$$

Moments about the mean: (central moments)

For each positive integer n ,
($n = 1, 2, \dots$) the n^{th} central moment

of the discrete random variable X is

$$\begin{aligned} \mu_n &= E(X - \bar{x})^n \\ &= \sum (x_i - \bar{x})^n p_i \end{aligned}$$

First Moment about the mean:

$$\begin{aligned} \mu_1 &= E(X - \bar{x})^1 = \sum (x_i - \bar{x})^1 p_i \\ &= \sum x_i p_i - \bar{x} \sum p_i \\ &= \sum x_i p_i - \bar{x} \quad (\because \sum p_i = 1) \\ &= E(X) - E(X) \end{aligned}$$

$$\boxed{\mu_1 = 0}$$

The algebraic sum of the deviations about the arithmetic mean is always zero.

Mean deviation about the mean is

$$M.D. = \sum |x_i - \text{mean}| p_i$$

Second Moment about the mean:

$$M_2 = E(X - \bar{X})^2$$

$$= E(X^2 + \bar{X}^2 - 2X\bar{X})$$

$$= E(X^2) - \bar{X}^2 - 2\bar{X}E(X) \quad [\because \bar{X} \text{ is a constant}]$$

$$= E(X^2) - (E(X))^2 - 2E(X)E(X)$$

$$M_2 = E(X^2) - (E(X))^2$$

Second moment about the mean is called the variance of the random variable X .

$$\begin{aligned} M_2 = \text{Var}(X) &= E(X - \bar{X})^2 \\ &= E(X^2) - (E(X))^2 \end{aligned}$$

Result:

1. $\text{Var}(X \pm c) = \text{Var } X$ where c is a Constant.

$\therefore$ Variance is independent of change of origin.

$$2. \text{Var}(aX) = a^2 \text{Var}(X)$$

Change of scale affects the Variance.

$$3. \text{Var}(c) = 0 \text{ where } c \text{ is a Constant.}$$

① Two unbiased dice are thrown together at random. Find the expected value of the total number of points shown up.

Sol:

Let X be the random variable which represents the sum of the numbers shown in the two dice.

If both show one then the sum is 2. If both show six then the sum is 12.

The random variable X can take values from 2 to 12.

(1,1)

(1,2) (2,1)

(1,3) (2,2) (3,1)

(1,4) (2,3) (3,2) (4,1)

(1,5) (2,4) (3,3) (4,2) (5,1)

(1,6) (2,5) (3,4) (4,3) (5,2) (6,1)

(2,6) (3,5) (4,4) (5,3) (6,2)

(3,6) (4,5) (5,4) (6,3)

(4,6) (5,5) (6,4)

(5,6) (6,5)

(6,6)

∴ The probability distribution is given by

X	2	3	4	5	6	7	8	9	10	11	12
$P(X=x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

$$E(X) = \sum p_i x_i = \sum x_i p_i$$

$$\begin{aligned} &= (2 \times \frac{1}{36}) + (3 \times \frac{2}{36}) + (4 \times \frac{3}{36}) + \\ &\quad (5 \times \frac{4}{36}) + (6 \times \frac{5}{36}) + (7 \times \frac{6}{36}) + \\ &\quad (8 \times \frac{5}{36}) + (9 \times \frac{4}{36}) + (10 \times \frac{3}{36}) + (11 \times \frac{2}{36}) \\ &\quad + (12 \times \frac{1}{36}) \end{aligned}$$

$$= \frac{252}{36}$$

$$= 7.$$

② The probability of success of an event is p and that of failure is q . Find the expected number of trials to get a first success.

Sol: Let X be the random variable denoting 'Number of trials to get a first success'.

The success can occur in the 1st trial.

$\therefore$ The probability of success in the 1st trial is p .

The success in the 2nd trial means failure in the 1st trial.

$\therefore$ probability is qP .

Success in the 3rd trial means failure in the first two trials.

$\therefore$ probability of success in the 3rd trial is q^2P .

As it goes on, the success may occur in the n th trial which means the first $(n-1)$ trials are failures.

$\therefore$ probability = $q^{n-1}P$.

$\therefore$ The probability distribution is as follows:

X	1	2	3	...	n
$P(X)$	P	qP	q^2P	...	$q^{n-1}P$

$$\therefore E(x) = \sum p_i x_i$$

$$= 1 \cdot p + 2qp + 3q^2p + \dots + nq^{n-1}p$$

$$= p(1 + 2q + 3q^2 + \dots + nq^{n-1})$$

$$= p(1-q)^{-2}$$

$$= p(p)^{-2}$$

$$= \frac{p}{p^2} = \frac{1}{p}$$

- ③ A game is played with a single fair die, A player wins Rs. 20 if a 2 turns up, Rs. 40 if a 4 turns up, while loses Rs. 30 if a 6 turns up. while he neither wins nor loses if any other face turns up. Find the expected sum of money he can win.

Sol: Let X be the random variable denoting the amount he can win.

The possible values of X are 20, 40, -30 and 0.

$$P(X=20) = P(\text{getting 2}) = \frac{1}{6}$$

$$P(X=40) = P(\text{getting 4}) = \frac{1}{6}$$

$$P(X=-30) = P(\text{getting 6}) = \frac{1}{6}$$

The remaining probability is $\frac{1}{2} = \frac{3}{6}$ for face 1, 3, 5.

X	20	40	-30	0
$P(X)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{2}$

$$\text{Mean } E(X) = \sum p_i x_i$$

$$= 20\left(\frac{1}{6}\right) + 40\left(\frac{1}{6}\right) + (-30)\left(\frac{1}{6}\right) + 0\left(\frac{1}{2}\right)$$

$$= 5.$$

Expected sum of money he can win = Rs. 5.

Expectation of a continuous random variable:

Let X be a Continuous random variable with probability density function $f(x)$. Then the mathematical expectation of X is defined as

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx.$$

Note:

1. $E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx.$

2. $E(c) = c$ where c is a Constant.

3. $E(ax \pm b) = aE(x) \pm b.$

Moments:

r^{th} Moment about origin:

$$\mu_r' = E(x^r) = \int_{-\infty}^{\infty} x^r f(x) dx.$$

First Moment about origin:

$$\mu_1' = E(x) = \int_{-\infty}^{\infty} x f(x) dx$$

Second Moment about origin:

$$\mu_2' = E(x^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$$

Note:

$$\text{Mean} = \bar{x} = \mu_1' = E(x).$$

rth Moment about mean:

rth Moment about mean is denoted by M_r and it is defined by

$$M_r = \int_{-\infty}^{\infty} (x - \bar{x})^r f(x) dx.$$

First moment about mean:

$$M_1 = \int_{-\infty}^{\infty} (x - \bar{x}) f(x) dx$$

$$= \int_{-\infty}^{\infty} x f(x) dx - \int_{-\infty}^{\infty} \bar{x} f(x) dx$$

$$= \bar{x} - \bar{x} \int_{-\infty}^{\infty} f(x) dx$$

$$= \bar{x} - \bar{x}$$

$$(\because \int_{-\infty}^{\infty} f(x) dx = 1)$$

$$\boxed{M_1 = 0}$$

Second moment about mean:

$$M_2 = \int_{-\infty}^{\infty} (x - \bar{x})^2 f(x) dx$$

$$\mu_2 = E[(x - \bar{x})^2]$$

$$\mu_2 = E(x^2) - (E(x))^2$$

$$\begin{aligned} \text{Variance} = \mu_2 &= \mu_2' - (\mu_1')^2 \\ &= E(x^2) - (E(x))^2 \end{aligned}$$

Moment about any point 'A':

$$\mu_r' = \int_{-\infty}^{\infty} (x - A)^r f(x) dx$$

Note:

$$\text{Mean} = E(x) = \int_{-\infty}^{\infty} x f(x) dx$$

① In a continuous distribution, the pdf of X is

$$f(x) = \begin{cases} \frac{3}{4}x(2-x), & 0 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$

Find the mean and the variance of the distribution.

Sol: $E(X) = \int_{-\infty}^{\infty} x f(x) dx$

$$= \int_0^2 x \frac{3x(2-x)}{4} dx$$

$$= \frac{3}{4} \int_0^2 (2x^2 - x^3) dx$$

$$= \frac{3}{4} \left(\frac{2x^3}{3} - \frac{x^4}{4} \right)_0^2$$

$$= \frac{3}{4} \left(\frac{16}{3} - \frac{16}{4} \right)$$

$\text{mean} = 1$

$$E(x^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$= \int_0^2 x^2 \frac{3x}{4} (2-x) dx$$

$$= \frac{3}{4} \left(2 \frac{x^4}{4} - \frac{x^5}{5} \right)_0^2$$

$$E(x^2) = \frac{3}{4} \left(\frac{16}{2} - \frac{32}{5} \right) = \frac{6}{5}$$

$$\text{Variance} = E(x^2) - (E(x))^2$$

$$= \frac{6}{5} - 1 = \frac{1}{5}$$

② Find the mean and variance of the distribution $f(x) = \begin{cases} 3e^{-3x}, & 0 < x < \infty \\ 0, & \text{elsewhere} \end{cases}$

Sol:

$$E(x) = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \int_0^{\infty} x (3e^{-3x}) dx$$

$$= 3 \int_0^{\infty} x e^{-3x} dx$$

$$= 3 \frac{1!}{3^2}$$

$$= \frac{1}{3}$$

$$\int_0^{\infty} x^n e^{-\alpha x} dx = \frac{n!}{\alpha^{n+1}}$$

Where n is a positive integer

$$E(x^2) = \int_0^{\infty} x^2 (3e^{-3x}) dx$$

$$= 3 \int_0^{\infty} x^2 e^{-3x} dx$$

$$= 3 \frac{2!}{3^3} = \frac{2}{9}$$

$$\text{Var}(x) = E(x^2) - (E(x))^2$$

$$= \frac{2}{9} - \left(\frac{1}{3}\right)^2$$

$$= \frac{2}{9} - \frac{1}{9}$$

$$\text{Var}(x) = \frac{1}{9}$$

$$\text{Mean} = \frac{1}{3} ; \text{Variance} = \frac{1}{9}$$

Two dimensional random Variable:

Let S be the sample space associated with a given random experiment. A real-valued function defined on S and taking values in $\mathbb{R} (-\infty, \infty)$ is called a one-dimensional random variable. If the function values are ordered pairs of real numbers (i.e. vectors in two-space) the function is said to be a two-dimensional random variable. More generally, an n -dimensional random variable is simply a function whose domain is S and whose range is a collection of n -tuples of real numbers (vectors in n -space).

Two dimensional random variable (Bi-variate random variable):

So far we have considered only the one-dimensional random variable, i.e. we have considered such random experiments, the outcome of which had only one characteristic and hence, was assigned a single real value. In many situations, we will be interested in recording two or more characteristics (numerically) of the outcome of a random experiment. For example, both voltage and current might be of interest in a certain experiment.

Also To know the health condition of a person, doctors measure many parameters like, height, weight, blood pressure, sugar level etc.

Definition:

Let S be the sample space associated with a random experiment E . Let $X = X(\omega)$ & $Y = Y(\omega)$ be two functions each assigning a real number to each outcomes $\omega \in S$. Then (X, Y) is called a two dimensional random variable.

Two dimensional discrete random variable:

If the possible values of (X, Y) are finite or countably infinite, then (X, Y) is called a two-dimensional discrete random variable.

When (X, Y) is a two-dimensional random variable, the possible values

of (x, y) may be represented as
 $(x_i, y_j), i=1, 2, \dots, m, \dots; j=1, 2, \dots, n, \dots$

Two dimensional continuous random variable:

If (x, y) can assume all values in a specified region R in the xy -plane, then (x, y) is called a two-dimensional continuous random variable.

Example:

Consider the experiment of tossing a coin twice. The sample space is

$$S = \{HH, HT, TH, TT\}$$

Let X denotes the number of heads obtained in the first toss and Y denotes the number of heads in the second toss.

Then

ξ	HH	HT	TH	TT
$X(\xi)$	1	1	0	0
$Y(\xi)$	1	0	1	0

(X, Y) is a two-dimensional random variable. The range space of (X, Y) is $\{(1, 1), (1, 0), (0, 1), (0, 0)\}$ which is finite and so (X, Y) is two-dimensional discrete random variables.

Joint Probability Mass function:

If (X, Y) is a two-dimensional discrete random variable such that

$$P(X=x_i, Y=y_j) = p_{ij} \quad (\text{or } p(x_i, y_j) \text{ or } p(x, y))$$

then p_{ij} is called the probability mass function of (X, Y) provided

$$(i) \quad p_{ij} \geq 0 \text{ for } i \text{ \& } j \quad p(x, y) \geq 0$$

$$(ii) \quad \sum_i \sum_j p_{ij} = 1.$$

Joint probability density function:

If (X, Y) is a two-dimensional continuous random variable then $f(x, y)$ satisfies the following

Conditions

$$i) \quad f(x, y) \geq 0 \text{ for all } (x, y) \in R$$

Where R is the range space

ii) $\iint_R f(x, y) dx dy = 1$ is called the joint pdf of (x, y) .

Note:

Moreover, if D is a subspace of the range space R , $P\{(x, y) \in D\}$ is defined as

$$P\{(x, y) \in D\} = \iint_D f(x, y) dx dy.$$

In particular

$$P\{a \leq x \leq b, c \leq y \leq d\} = \int_c^d \int_a^b f(x, y) dx dy$$

Cumulative distribution function (cdf)

(or) joint distribution function (or)

joint probability distribution:

If (x, y) is a two-dimensional random variable (discrete or continuous)

then $F(x, y) = P(X \leq x \text{ and } Y \leq y)$ (or)
 $P(-\infty < X \leq x, -\infty < Y \leq y)$ is called the
cdf of (X, Y) .

For discrete Case

$$F(x, y) = \sum_j \sum_i P_{ij}$$

$y_j \leq y; x_i \leq x$

For Continuous Case

$$F(x, y) = \int_{-\infty}^y \int_{-\infty}^x f(x, y) dx dy$$

Properties of $F(x, y)$:

1. $F(-\infty, y) = 0 = F(x, -\infty) = F(-\infty, -\infty)$
2. $F(\infty, \infty) = 1$
3. $0 \leq F(x, y) \leq 1$
4. $P(a < X < b, Y \leq y) = F(b, y) - F(a, y)$

$$5. P(X \leq x, C < Y < d) = F(x, d) - F(x, c)$$

$$6. P(a < X < b, C < Y < d) = F(b, d) - F(a, d) - F(b, c) + F(a, c)$$

7. At points of continuity of $f(x, y)$,

$$\frac{\partial^2 F}{\partial x \partial y} = f(x, y).$$

Marginal Probability Distribution:

$$P(X = x_i) = P((X = x_i \text{ and } Y = y_1) \text{ or } (X = x_i \text{ and } Y = y_2) \text{ or etc})$$

$$= p_{i1} + p_{i2} + \dots$$

$P(X = x_i) = \sum_j p_{ij}$ is called the marginal probability function of X . It is defined for $X = x_1, x_2, \dots$ and denoted as P_{i*} (or)

$$P_{i*} \quad \text{(or)} \quad P_X(x) = \sum_y P(x, y)$$

The collection of pairs $\{x_i, p_{i*}\}$, $i = 1, 2, 3, \dots$ is called the marginal probability distribution of X .

Similarly $P(Y = y_j) = \sum_i p_{ij} = p_{*j}$ (or) $P_{.j}$ is called the marginal probability function of Y . $P_Y(y) = \sum_x p(x, y)$

The collection of pairs $\{y_j, P_{*j}\}$, $j = 1, 2, \dots$ is called the marginal probability distribution of Y .

for Continuous Case,

$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$ is called the

marginal density of X .

$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$ is called the marginal density of Y .

Note:

$$P(a \leq x \leq b) = P(a \leq x \leq b, -\infty < y < \infty)$$

$$= \int_{-\infty}^{\infty} \int_a^b f(x, y) dx dy$$

$$= \int_a^b \left(\int_{-\infty}^{\infty} f(x, y) dy \right) dx$$

$$P(a \leq x \leq b) = \int_a^b f_x(x) dx$$

$$\text{III}^{\text{ly}} P(c \leq y \leq d) = \int_c^d f_y(y) dy$$

Conditional probability distribution:

$$P(X = x_i / Y = y_j) = \frac{P(X = x_i, Y = y_j)}{P(Y = y_j)} = \frac{p_{ij}}{p_{*j}}$$

$$P_{X/Y}(x/y) = \frac{p(x, y)}{p_Y(y)}$$

is called the conditional probability function of X , given that $Y = y_j$.

The collection of pairs $\{x_i, \frac{p_{ij}}{p_{i*}}\}$
 $i=1, 2, \dots$ is called the conditional
probability distribution of X given
 $Y = y_j$.

$$\text{III}^{\text{ly}} \quad P_{Y/X}(y/x) = \frac{p(x, y)}{P_X(x)}$$

$$P(Y = y_j \mid X = x_i) = \frac{P(X = x_i, Y = y_j)}{P(X = x_i)} = \frac{p_{ij}}{p_{i*}}$$

is called the conditional probability
function of Y , given that $X = x_i$.

The collection of pairs $\{y_j, \frac{p_{ij}}{p_{i*}}\}$
is called the conditional probability
distribution of Y , given $X = x_i$.

In Continuous Case,

$\frac{f(x,y)}{f_y(y)}$ is called the conditional density of X , given y and is denoted by $f(x/y)$ (or) $f_x^y(x/y)$.

||| $\frac{f(x,y)}{f_x(x)}$ is called the conditional density of Y , given x and is denoted by $f(y/x)$ (or) $f_{y/x}(y/x)$.

Independent random Variables:

If (X, Y) is a two-dimensional discrete random variable such that

$$P(X=x_i / Y=y_j) = P(X=x_i)$$

$$\text{i.e. } \frac{P_{ij}}{P_{*j}} = P_{i*} \quad P(X,Y) = P_X(X)P_Y(Y)$$

$$\text{i.e. } P_{ij} = P_{i*} P_{*j} \quad (\text{or}) \quad \text{for all } i, j$$

then X and Y are said to be independent random variables.

Similarly,

If (X, Y) is a two-dimensional continuous random variable such that $f(x, y) = f_X(x) f_Y(y)$ then X & Y are said to be independent random variables.

Note:

$$P_{Y/X}(y/x) = \frac{P(x,y)}{P_X(x)} = \frac{P_X(x) P_Y(y)}{P_X(x)} = P_Y(y)$$

$$P_{X/Y}(x/y) = \frac{P(x,y)}{P_Y(y)} = \frac{P_X(x) P_Y(y)}{P_Y(y)} = P_X(x)$$

Thus, in the case of independence,

marginal distributions $\equiv$ Conditional distributions.

$$\text{III}^{\text{ly}} \quad f(x/y) = f_x(x), \quad f(y/x) = f_y(y)$$

Thus, in the case of independence,

marginal distributions $\equiv$ Conditional distributions.

Expected Values of a two dimensional
Random Variable:

for discrete case

$$E(xy) = \sum_j \sum_i x_i y_j p_{ij}$$

for continuous case

$$E(xy) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x,y) dx dy$$

properties:

1. $E(x+y) = E(x) + E(y)$

2. $E(xy) \neq E(x)E(y)$ in general.

If x and y are independent random variables then $E(xy) = E(x)E(y)$.

Covariance:

If x and y are two random variables then Covariance between them is defined as

$$\text{Cov}(x, y) = E(xy) - E(x)E(y)$$

Note:

1. If x & y are independent then $E(xy) = E(x)E(y)$
 $\therefore \text{Cov}(x, y) = 0$

$$2. \text{Cov}(ax, by) = ab \text{Cov}(x, y)$$

$$3. \text{Cov}(x+a, y+b) = \text{Cov}(x, y)$$

$$4. \text{Cov}(ax+b, cy+d) = ac \text{Cov}(x, y)$$

$$5. \text{Var}(x_1 + x_2) = \text{Var}(x_1) + \text{Var}(x_2) + 2 \text{Cov}(x_1, x_2)$$

$$6. \text{Var}(x_1 - x_2) = \text{Var}(x_1) + \text{Var}(x_2) - 2 \text{Cov}(x_1, x_2)$$

Correlation:

Correlation is a Statistical tool that helps to measure and analyze the relationship between two variables.

Correlation is a measure of association between two variables.

The measure of correlation called the Correlation Coefficient.

Correlation Coefficient between the variables X & Y is defined as

$$r_{xy} = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y}$$

Note:

1. $-1 \leq r \leq 1$
2. $r = 1$ perfect positive correlation
3. $0 < r < 1$ positive correlation
4. $-1 < r < 0$ negative correlation
5. $r = -1$ perfect negative correlation
6. $r = 0$, no correlation.

① The joint probability mass function of (X, Y) is given by $P(X, Y) = K(2x + 3y)$, $x = 0, 1, 2$, $y = 1, 2, 3$. Find all the marginal & conditional probability distributions. Also find the probability distribution of $X + Y$. Also verify whether X & Y are independent. Find mean of X , mean of Y , $\text{Var}(X)$, $\text{Var}(Y)$, $\text{Cov}(X, Y)$, Coefficient of correlation between X & Y .

Solution:

The joint probability distribution of (X, Y) is given below.

$x \backslash y$	1	2	3
0	$3k$	$6k$	$9k$
1	$5k$	$8k$	$11k$
2	$7k$	$10k$	$13k$

To find k:

$$\sum_i \sum_j p_{ij} = 1$$

$$(or) \sum_{j=1}^3 \sum_{i=0}^2 p(x_i, x_j) = 1$$

$$\Rightarrow 72k = 1$$

$$\Rightarrow \boxed{k = \frac{1}{72}}$$

Marginal probability distribution of x:

$$p_{i*} = P(X = x_i) = \sum_{j=1}^3 p_{ij}$$

$$\begin{aligned} i) P(X=0) &= P_{0*} = p_{01} + p_{02} + p_{03} \\ &= P(X=0, Y=1) + P(X=0, Y=2) + P(X=0, Y=3) \\ &= 3k + 6k + 9k = 18k = \frac{18}{72} \end{aligned}$$

$$\begin{aligned}
 \text{ii) } P(X=1) &= p_{1*} = p_{11} + p_{12} + p_{13} \\
 &= P(X=1, Y=1) + P(X=1, Y=2) + P(X=1, Y=3) \\
 &= 5k + 8k + 11k \\
 &= \frac{24}{72}
 \end{aligned}$$

$$\begin{aligned}
 \text{iii) } P(X=2) &= p_{2*} = p_{21} + p_{22} + p_{23} \\
 &= 7k + 10k + 13k = \frac{30}{72}
 \end{aligned}$$

Marginal probability distribution of X is $\{x_i, p_{i*}\}$.

$x=x_i$	$P_{i*} = \sum_{j=1}^3 p_{ij}$
0	$\frac{18}{72}$
1	$\frac{24}{72}$
2	$\frac{30}{72}$
	Total 1

Marginal probability distribution of Y:

$$p_{*j} = P(Y=y_j) = \sum_{i=0}^2 p_{ij}$$

$$i) P(Y=1) = p_{*1} = p_{01} + p_{11} + p_{21} = 3k + 5k + 7k = \frac{15}{72}$$

$$ii) P(Y=2) = p_{*2} = p_{02} + p_{12} + p_{22} = 6k + 8k + 10k = \frac{24}{72}$$

$$iii) P(Y=3) = p_{*3} = p_{03} + p_{13} + p_{23} = 9k + 11k + 13k = \frac{33}{72}$$

Marginal probability distribution of Y is

$\{y_j, p_{*j}\}$

$Y = y_j$	$p_{*j} = \sum_{i=0}^2 p_{ij}$
1	$\frac{15}{72}$
2	$\frac{24}{72}$
3	$\frac{33}{72}$
Total	1

Conditional probability distributions:

Conditional probability distribution of $X=x_i$ given that $Y=y_j$ is

$$P(X=x_i / Y=y_j) = \frac{P(X=x_i, Y=y_j)}{P(Y=y_j)} = \frac{p_{ij}}{p_{+j}}$$

i) Conditional probability of X given
 $Y=1$:

$$P(X=0 / Y=1) = \frac{P(X=0, Y=1)}{P(Y=1)} = \frac{p_{01}}{P(Y=1)}$$

$$= \frac{3/72}{15/72} = \frac{3}{15} = \frac{1}{5}$$

$$P(X=1 / Y=1) = \frac{P(X=1, Y=1)}{P(Y=1)} = \frac{p_{11}}{P(Y=1)} = \frac{5/72}{15/72} = \frac{1}{3}$$

$$P(X=2/Y=1) = \frac{p_{21}}{P(Y=1)} = \frac{7/42}{15/42} = \frac{7}{15}$$

Conditional probability distribution

X given $Y=1$:

$X=x_i$	p_{i1}/p_{*1}
0	$1/5$
1	$1/3$
2	$7/15$
Total	1

Conditional probability of X given $Y=2$:

$$P(X=0/Y=2) = \frac{P(X=0, Y=2)}{P(Y=2)} = \frac{6/72}{24/72} = \frac{6}{24} = \frac{1}{4}$$

$$P(X=1/Y=2) = \frac{P(X=1, Y=2)}{P(Y=2)} = \frac{8/72}{24/72} = \frac{8}{24} = \frac{1}{3}$$

$$P(X=2/Y=2) = \frac{P(X=2, Y=2)}{P(Y=2)} = \frac{\frac{10}{72}}{\frac{24}{72}} = \frac{10}{24} = \frac{5}{12}$$

Conditional probability distribution of
X given Y=2:

$X=x_i$	P_{i2}/p_{*2}
0	$\frac{1}{4}$
1	$\frac{1}{3}$
2	$\frac{5}{12}$
Total	1

Conditional probability of X given Y=3:

$$P(X=0/Y=3) = \frac{P(X=0, Y=3)}{P(Y=3)} = \frac{\frac{9}{72}}{\frac{33}{72}} = \frac{9}{33} = \frac{3}{11}$$

$$P(X=1/Y=3) = \frac{\frac{11}{72}}{\frac{33}{72}} = \frac{11}{33} = \frac{1}{3}$$

$$P(X=2/Y=3) = \frac{\frac{13}{72}}{\frac{33}{72}} = \frac{13}{33}$$

Conditional probability distribution of X given $Y=3$:

$X=x_i$	P_{i3}/P_{*3}
0	$\frac{3}{11}$
1	$\frac{1}{3}$
2	$\frac{13}{33}$
Total	1

Conditional probability of $Y=y_j$ given $X=x_i$ is

$$P(Y=y_j / X=x_i) = \frac{P(X=x_i, Y=y_j)}{P(X=x_i)} = \frac{P_{ij}}{P_{i*}}$$

Conditional probability of Y given $X=0$:

$$P(Y=1 / X=0) = \frac{P(X=0, Y=1)}{P(X=0)} = \frac{3k}{18k} = \frac{3}{18} = \frac{1}{6}$$

$$P(Y=2 / X=0) = \frac{6k}{18k} = \frac{1}{3}$$

$$P(Y=3 / X=0) = \frac{9k}{18k} = \frac{1}{2}$$

Conditional probability distribution of y

Given $X=0$:

$Y = y_i$	$p_{0j} / p_{0\cdot}$
1	$\frac{1}{6}$
2	$\frac{1}{3}$
3	$\frac{1}{2}$
Total	1

Conditional distribution of y given $X=1$:

$$P(Y=1/X=1) = \frac{P(X=1, Y=1)}{P(X=1)} = \frac{5k}{24k} = \frac{5}{24}$$

$$P(Y=2/X=1) = \frac{8k}{24k} = \frac{1}{3}$$

$$P(Y=3/X=1) = \frac{11k}{24k} = \frac{11}{24}$$

$Y = y_j$	P_{1j} / p_{1*}
1	$5/24$
2	$1/3$
3	$11/24$
Total	1

Conditional distribution of Y given $X=2$:

$$P(Y=1/X=2) = \frac{P(X=2, Y=1)}{P(X=2)} = \frac{7k}{30k} = \frac{7}{30}$$

$$P(Y=2/X=2) = \frac{10k}{30k} = \frac{1}{3}$$

$$P(Y=3/X=2) = \frac{13k}{30k} = \frac{13}{30}$$

$Y = y_j$	P_{2j} / p_{2*}
1	$7/30$
2	$1/3$
3	$13/30$
Total	1

Probability distribution of $(x+y)$:

$x+y$	$P(x+y)$
1	$p_{01} = \frac{3}{72}$
2	$p_{02} + p_{11} = \frac{11}{72}$
3	$p_{03} + p_{12} + p_{21} = \frac{24}{72}$
4	$p_{13} + p_{22} = \frac{21}{72}$
5	$p_{23} = \frac{13}{72}$

Verification for x & y are independent or not:

$p_{ij} = p_{i*} \times p_{*j}$ for all i & j .
if x & y are independent.

$$p_{11} = \frac{5}{72} = 0.0694 \quad p_{1*} = P(x=1) = \frac{24}{72}$$

$$p_{*1} = \frac{15}{72}$$

$$p_{1*} \times p_{*1} = \frac{24}{72} \times \frac{15}{72} = \frac{360}{5184} = 0.0694$$

$$1) p_{11} = p_{1*} \times p_{*1}$$

$$p_{22} = \frac{10}{72} = 0.13888$$

$$p_{2*} = \frac{30}{72}, \quad p_{*2} = \frac{24}{72}$$

$$p_{2*} \times p_{*2} = 0.13888$$

$$2) p_{22} = p_{2*} \times p_{*2}$$

$$p_{12} = p_{1*} \times p_{*2}$$

$$\frac{8}{72} = \frac{24}{72} \times \frac{24}{72}$$

$$0.1111 = 0.1111$$

$$3) p_{12} = p_{1*} \times p_{*2}$$

$$4) p_{13} = p_{1*} \times p_{*3}$$

$$\frac{11}{72} = \frac{24}{72} \times \frac{33}{72}$$

$$0.1527 = 0.1527$$

$$5) p_{01} = p_{0*} \times p_{*1}$$

$$\frac{3}{72} = \frac{18}{72} \times \frac{15}{72}$$

$$0.04167 \neq 0.05208$$

$\therefore X$ & Y are not independent.

Mean of X:

$$E(X) = \sum_{i=0}^2 x_i p_i$$

$$= 0 \times P(X=1) + 1 \times P(X=2) + 2 \times P(X=3)$$

$$= 0 \times \frac{18}{72} + 1 \times \frac{24}{72} + 2 \times \frac{30}{72}$$

$$= \frac{24}{72} + \frac{60}{72} = \frac{84}{72} = 1.167$$

$$E(Y) = \sum_{j=1}^3 y_j p_j$$

$$= 1 \times P(Y=1) + 2 \times P(Y=2) + 3 \times P(X=3)$$

$$= 1 \times \frac{15}{72} + 2 \times \frac{24}{72} + 3 \times \frac{33}{72}$$

$$= \frac{15}{72} + \frac{48}{72} + \frac{99}{72} = \frac{162}{72} = 2.25$$

$$E(XY) = \sum_j \sum_i x_i y_j p_{ij}$$

$$= 0 \times 1 \times \frac{3}{72} + 0 \times 2 \times \frac{6}{72} + 0 \times 3 \times \frac{9}{72} +$$

$$1 \times 1 \times \frac{5}{72} + 1 \times 2 \times \frac{8}{72} + 1 \times 3 \times \frac{11}{72} +$$

$$2 \times 1 \times \frac{7}{72} + 2 \times 2 \times \frac{10}{72} + 2 \times 3 \times \frac{13}{72}$$

$$= 0 + 0 + 0 + \frac{5}{72} + \frac{16}{72} + \frac{33}{72} + \frac{14}{72} + \frac{40}{72} + \frac{78}{72}$$

$$= \frac{186}{72}$$

$$E(xy) = 2.583$$

$$E(xy) \neq E(x) E(y) \quad (\because E(x) E(y) = 2.62575)$$

$\therefore X$ & Y are not independent.

$$\text{Cov}(X, Y) = E(xy) - E(x) E(y)$$

$$= \frac{186}{72} - \frac{84}{72} \times \frac{162}{72}$$

$$\text{Cov}(X, Y) = -0.04275$$

$$\gamma(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Where $\sigma_X = \sqrt{\text{var}(X)}$, $\sigma_Y = \sqrt{\text{var}(Y)}$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$\text{Var}(Y) = E(Y^2) - (E(Y))^2$$

$$E(x^2) = \sum_{i=0}^2 x_i^2 p_i$$

$$= 0 \times \frac{18}{72} + 1^2 \times \frac{24}{72} + 2^2 \times \frac{30}{72}$$

$$= \frac{24}{72} + \frac{120}{72} = \frac{144}{72} = 2$$

$$E(y^2) = \sum_{j=1}^3 y_j^2 p_j$$

$$= 1^2 \times \frac{15}{72} + 2^2 \times \frac{24}{72} + 3^2 \times \frac{33}{72}$$

$$= \frac{15}{72} + \frac{96}{72} + \frac{297}{72} = \frac{408}{72} = 5.667$$

$$\text{var}(x) = E(x^2) - (E(x))^2$$

$$= 2 - (1.167)^2$$

$$\text{var}(x) = 2 - 1.361889 = 0.638111$$

$$\sigma_x = 0.79882$$

$$\text{var}(y) = E(y^2) - (E(y))^2$$

$$= 5.667 - (2.25)^2$$

$$= 5.667 - 5.0625$$

$$\text{var}(y) = 0.6045$$

$$\sigma_y = 0.7775$$

$$\sigma_x \sigma_y = 0.62108$$

$$r_{xy} = \frac{-0.04275}{0.62108}$$

$$r_{xy} = -0.06883$$

② The joint pdf of a 2-dimensional random variable (x, y) is given by

$$f(x, y) = \begin{cases} xy^2 + \frac{x}{8} & , 0 \leq x \leq 2, \\ & 0 \leq y \leq 1 \\ 0 & , \text{otherwise} \end{cases}$$

then compute

i) $P(x > 1)$, $P(y < \frac{1}{2})$, $P(x > 1 / y < \frac{1}{2})$
 $P(y < \frac{1}{2} / x > 1)$, $P(x < y)$.

ii) $P(x + y \leq 1)$.

Solution:

$$P(X > 1) = \int_{y=0}^1 \int_{x=1}^2 f(x, y) dx dy$$

$$= \int_0^1 \int_1^2 (xy^2 + \frac{x^2}{8}) dx dy$$

$$= \int_0^1 \left(\frac{x^2}{2} y^2 + \frac{x^3}{24} \right) \Big|_1^2 dy$$

$$= \int_0^1 \left[\left(2y^2 + \frac{1}{3} \right) - \left(\frac{y^2}{2} + \frac{1}{24} \right) \right] dy$$

$$= \int_0^1 \left(\frac{3y^2}{2} + \frac{7}{24} \right) dy$$

$$= \left(\frac{y^3}{2} + \frac{7}{24} y \right) \Big|_0^1$$

$$P(X > 1) = \frac{1}{2} + \frac{7}{24} = \frac{12+7}{24} = \frac{19}{24}$$

$$P(Y < \frac{1}{2}) = \int_{y=0}^{\frac{1}{2}} \int_{x=0}^2 (xy^2 + \frac{x^2}{8}) dx dy = \frac{1}{4}$$

$$P(x > 1, y < \frac{1}{2}) = \int_{y=0}^{\frac{1}{2}} \int_{x=1}^2 (xy^2 + \frac{x^2}{8}) dx dy = \frac{5}{24}$$

$$P(x > 1 / y < \frac{1}{2}) = \frac{P(x > 1, y < \frac{1}{2})}{P(y < \frac{1}{2})}$$

$$= \frac{\frac{5}{24}}{\frac{1}{4}} = \frac{5}{6}$$

$$P(y < \frac{1}{2} / x > 1) = \frac{P(x > 1, y < \frac{1}{2})}{P(x > 1)}$$

$$= \frac{\frac{5}{24}}{\frac{19}{24}} = \frac{5}{19}$$

$$P(x < y) = \int_0^1 \int_0^y (xy^2 + \frac{x^2}{8}) dx dy$$

$$= \frac{53}{480}$$

$$P(x + y \leq 1) = \int_{y=0}^1 \int_{x=0}^{1-y} (xy^2 + \frac{x^2}{8}) dx dy$$

$$= \frac{13}{480}$$

③ The joint pdf of (x, y) is given by $f(x, y) = kxy e^{-(x^2+y^2)}$, $x > 0, y > 0$. Find the value of k and verify whether X & Y are independent.

Solution:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

$$\Rightarrow \int_0^{\infty} \int_0^{\infty} kxy e^{-(x^2+y^2)} dx dy = 1$$

$$\Rightarrow k \int_0^{\infty} x e^{-x^2} dx \int_0^{\infty} y e^{-y^2} dy = 1$$

$$\Rightarrow k \cdot \frac{1}{2} \cdot \frac{1}{2} = 1$$

$$\Rightarrow \boxed{k = 4}$$

$$\begin{aligned} \text{put } t &= x^2 \\ dt &= 2x dx \\ \frac{dt}{2} &= x dx \end{aligned}$$

$$\begin{aligned} x &: 0 \rightarrow \infty \\ t &: 0 \rightarrow \infty \end{aligned}$$

$$\int_0^{\infty} x e^{-x^2} dx$$

$$= \frac{1}{2} \int_0^{\infty} e^{-t} dt$$

$$= \frac{1}{2} \left(\frac{e^{-t}}{-1} \right)_0^{\infty}$$

$$= -\frac{1}{2} (e^{-\infty} - e^0)$$

$$= -\frac{1}{2} (0 - 1)$$

$$= \frac{1}{2}$$

$$f(x) = \int_0^{\infty} f(x, y) dy$$

$$= 4 \int_0^{\infty} xy e^{-x^2-y^2} dx dy$$

$$= 4x e^{-x^2} \int_0^{\infty} y e^{-y^2} dy$$

$$f(x) = \cancel{x} e^{-x^2} \frac{1}{2} = 2x e^{-x^2}$$

$$\text{III}^{\text{ly}} f(y) = 2y e^{-y^2}$$

$$f(x, y) = 4xy e^{-x^2-y^2}$$

$$f(x) f(y) = 4xy e^{-x^2-y^2}$$

$$f(x, y) = f(x) f(y)$$

$\therefore X$ & Y are independent.

Moment generating function (mgf)

The mgf of a random variable X (about origin) whose probability function is given by

$$M_X(t) = E(e^{tx})$$

$$= \begin{cases} \sum_x e^{tx} p(x) & \text{for discrete case} \\ \int_{-\infty}^{\infty} e^{tx} f(x) dx & \text{for continuous case} \end{cases}$$

Where t is real parameter and the integration or summation being extended to the entire range of x .

To find r^{th} moment of X about origin:

we know that

$$M_x(t) = E(e^{tx})$$

$$= E\left(1 + \frac{tx}{1!} + \frac{(tx)^2}{2!} + \frac{(tx)^3}{3!} + \dots + \frac{(tx)^r}{r!} + \dots\right)$$

$$= 1 + t E(x) + \frac{t^2}{2!} E(x^2) + \dots + \frac{t^r}{r!} E(x^r) + \dots$$

$$M_x(t) = 1 + t M_1' + \frac{t^2}{2!} M_2' + \frac{t^3}{3!} M_3' + \dots + \frac{t^r}{r!} M_r' \quad \text{--- (1)}$$

$$= \sum_{r=0}^{\infty} \frac{t^r}{r!} M_r'$$

$$\therefore M_r' = E(x^r)$$

This gives the m.g.f. in terms of moments.
Thus the coefficient of $\frac{t^r}{r!}$ in $M_x(t)$ gives the r^{th} moment of the random variable X about origin (M_r').

Note:

Since $M_X(t)$ generates moments, it is known as m.g.f.

Note:

Diff (1) w.r.t. 't' we get

$$M_X'(t) = M_1' + \frac{2t}{2!} M_2' + \frac{3t^2}{3!} M_3' + \dots \rightarrow (2)$$

$$M_X'(0) = M_1'$$

The first moment about origin is given by $M_1' = M_X'(0) = \bar{x} = \text{mean}.$

Diff (2) w.r.t. 't',

$$M_X''(t) = M_2' + t M_3' + \dots$$

$$M_X''(0) = M_2'$$

Second moment about origin is given by $M_2' = M_X''(0).$

In general, we get

$$M_1' = \left(\frac{d}{dt} M_x(t) \right)_{t=0} = 0.$$

Note:

The mgf of X about the point $x=a$ is defined by

$$M_x(t) = E(e^{t(x-a)})$$

$$= E\left(1 + t(x-a) + \frac{t^2}{2!} (x-a)^2 + \dots\right)$$

$$+ \frac{t^3}{3!} (x-a)^3 + \dots$$

$$= 1 + t E(x-a) + \frac{t^2}{2!} E(x-a)^2 + \dots$$

$$+ \frac{t^3}{3!} E(x-a)^3 + \dots$$

$$M_x(t) = 1 + t \mu_1' + \frac{t^2}{2!} \mu_2' + \dots + \frac{t^r}{r!} \mu_r' + \dots$$

Where $\mu_r' = E((x-a)^r)$ which gives the r^{th} moment about the point $x=a$.

Results:

$$M_{cx}(t) = E(e^{tcx}) \longrightarrow \textcircled{1}$$

$$M_x(ct) = E(e^{ctx}) \longrightarrow \textcircled{2}$$

From (1) & (2)

$$M_{cx}(t) = M_x(ct)$$

Theorem:

If $X_1, X_2, \dots, X_n$ are independent

random variables then

$$M_{X_1+X_2+\dots+X_n}(t) = M_{X_1}(t) M_{X_2}(t) \dots M_{X_n}(t)$$

Characteristic function:

In few case the m.g.f does not exist. In this case we use characteristic functions which is more serviceable function than the m.g.f.

The characteristic function of the random variable X whose pdf is defined as follows

$$\varphi_x(t) = E(e^{itx}) = \int_{-\infty}^{\infty} e^{itx} f(x) dx$$

for continuous case

$$= \sum_x e^{itx} p(x) \text{ for discrete case}$$

Note:

we denote $\varphi_x(t)$ by $\varphi(t)$.

Property:

1. $\varphi_x(0) = 1$ (or) $\varphi(0) = 1$

2. $|\varphi(t)| \leq 1$

3. $\varphi(-t) = \overline{\varphi(t)}$

Theorem:

If X_1 & X_2 are two independent random variables then

$$\varphi_{X_1+X_2}(t) = \varphi_{X_1}(t) \times \varphi_{X_2}(t).$$

In general,

$$\varphi_{X_1+X_2+\dots+X_n}(t) = \varphi_{X_1}(t) \varphi_{X_2}(t) \dots \varphi_{X_n}(t).$$

Note:

Converse is not true.

① Find the mgf of the random variable with probability law

$$P(X=x) = q^{x-1} p, \quad x=1, 2, 3, \dots$$

Find the mean & Variance.

Solution:

$$M_x(t) = E(e^{tx})$$

$$= \sum_{x=1}^{\infty} e^{tx} p(x)$$

$$= \sum_{x=1}^{\infty} e^{tx} q^{x-1} p$$

$$= \sum_{x=1}^{\infty} e^{tx} q^x q^{-1} p$$

$$= \frac{p}{q} \sum_{x=1}^{\infty} (e^t q)^x$$

$$= \frac{p}{q} q e^t \sum_{x=1}^{\infty} (e^t q)^{x-1}$$

$$= p e^t (1 + q e^t + (q e^t)^2 + \dots)$$

$$= p e^t (1 - q e^t)^{-1} \quad (\because (1-x)^{-1} = 1 + x + x^2 + \dots)$$

$$M_x(t) = \frac{p e^t}{1 - q e^t}$$

$$M'_x(t) = \frac{p e^t (1 - q e^t) - p e^t (-q e^t)}{(1 - q e^t)^2}$$

$$= \frac{pe^t - pqe^{2t} + pqe^{2t}}{(1-qe^t)^2}$$

$$M_x'(t) = \frac{pe^t}{(1-qe^t)^2}$$

$$M_x'(0) = \frac{p}{(1-q)^2} = \frac{p}{p^2} = \frac{1}{p}$$

$$M_1' = M_x'(0) = \frac{1}{p} = \text{mean (about origin)}$$

$$M_x''(t) = \frac{(1-qe^t)^2 pe^t - pe^t \cdot 2(1-qe^t)(-qe^t)}{(1-qe^t)^4}$$

$$= \frac{(1-qe^t) (pe^t - pqe^{2t} + 2pqe^{2t})}{(1-qe^t)^3}$$

$$M_x''(t) = \frac{pe^t + pqe^{2t}}{(1-qe^t)^3}$$

$$M_x''(0) = \frac{p+pq}{(1-q)^3} = \frac{p+pq}{p^3} = \frac{1+q}{p^2} = M_2'$$

$$\text{Variance} = \mu_2' - (\mu_1')^2$$

$$= \frac{1+2}{p^2} - \frac{1}{p^2}$$

$$\text{Variance} = \frac{1+2-1}{p^2} = \frac{2}{p^2}$$

② Find the characteristic function of the distribution with pdf $f(x) = \frac{\alpha}{2} e^{-\alpha|x|}$, $-\infty < x < \infty$. Hence find its mean & variance.

Sol:

$$\phi(t) = \int_{-\infty}^{\infty} f(x) e^{itx} dx$$

$$= \frac{\alpha}{2} \int_{-\infty}^{\infty} e^{itx} e^{-\alpha|x|} dx$$

$$= \frac{\alpha}{2} \left[\int_{-\infty}^0 e^{\alpha x} e^{itx} dx + \int_0^{\infty} e^{-\alpha x} e^{itx} dx \right]$$

$$= \frac{\alpha}{2} \left[\int_{-\infty}^0 e^{(\alpha+it)x} dx + \int_0^{\infty} e^{(-\alpha+it)x} dx \right]$$

$$= \frac{\alpha}{2} \left[\frac{1}{\alpha+it} + \frac{1}{\alpha-it} \right]$$

$$\phi(t) = \frac{\alpha^2}{\alpha^2 + t^2} = \frac{1}{1 + \frac{t^2}{\alpha^2}} = \left(1 + \frac{t^2}{\alpha^2}\right)^{-1} = 1 - \frac{t^2}{\alpha^2} + \frac{t^4}{\alpha^4} - \dots$$

$$\phi'(t) = -\frac{2t}{\alpha^2} + \frac{4t^3}{\alpha^4} - \dots$$

$$\mu_1' = (\phi'(t))_{t=0} = 0 = \text{mean}$$

$$\phi''(t) = -\frac{2}{\alpha^2} + \frac{12t^2}{\alpha^4}$$

$$(\phi''(t))_{t=0} = -\frac{2}{\alpha^2} = \mu_2'$$

$$\begin{aligned} \text{Variance} &= \mu_2' - (\mu_1')^2 \\ &= -\frac{2}{\alpha^2} \end{aligned}$$